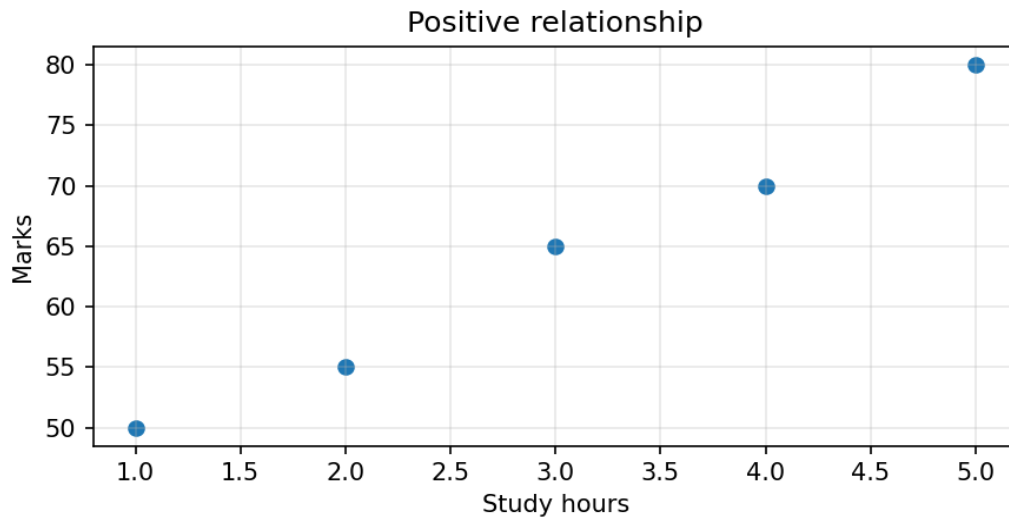


DAY 22

CORRELATION & COVARIANCE

Visual, step-by-step Data Science study guide

Prepared by Muhammad Ahmad



[Explanations](#) • [Visual examples](#) • [Python code](#) • [Practice tasks](#) • [Full solutions](#) • [Mini-project](#)

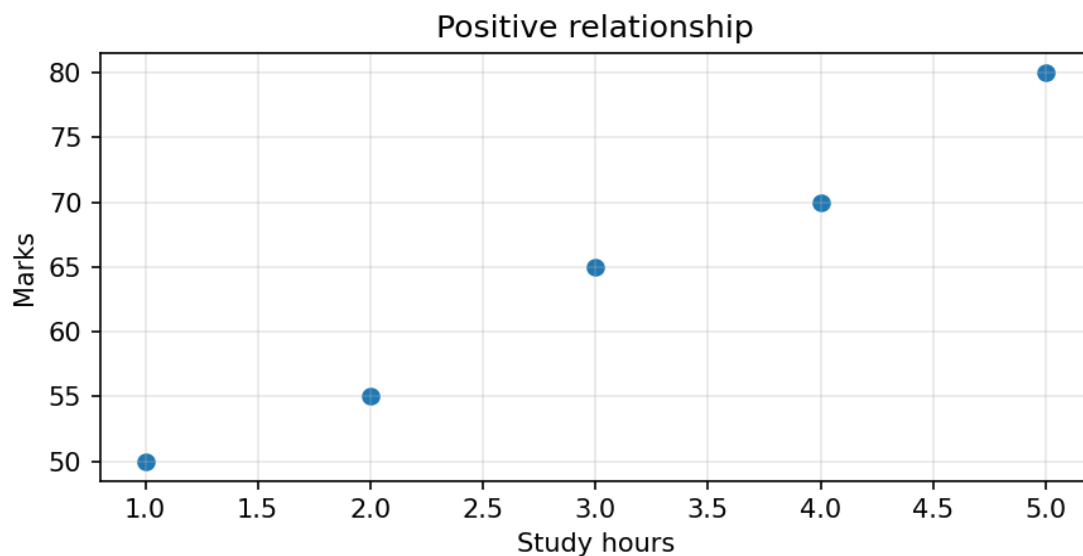
Learning objectives

- Explain covariance and correlation in simple language.
- Recognize positive, negative, and weak/no linear relationships in scatter plots.
- Calculate Pearson and Spearman correlation using Python.
- Create correlation matrices and heatmaps with Pandas and Matplotlib.
- Interpret relationships cautiously and explain why correlation does not prove causation.

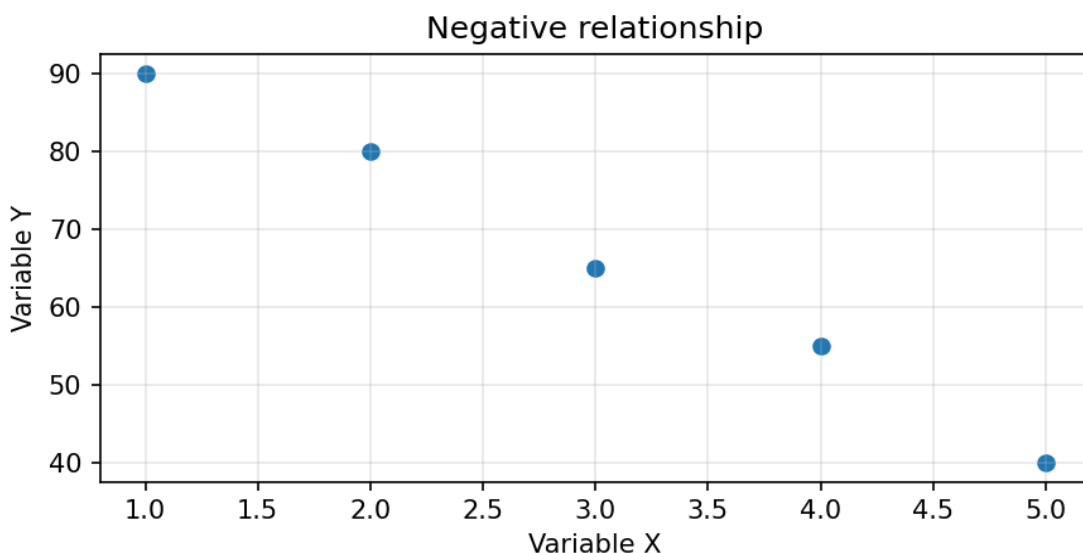
1. What is correlation?

Correlation describes the direction and strength of association between two variables. For example, we can investigate whether study hours and exam marks tend to increase together. It summarizes observed patterns, but does not by itself prove that one variable causes the other.

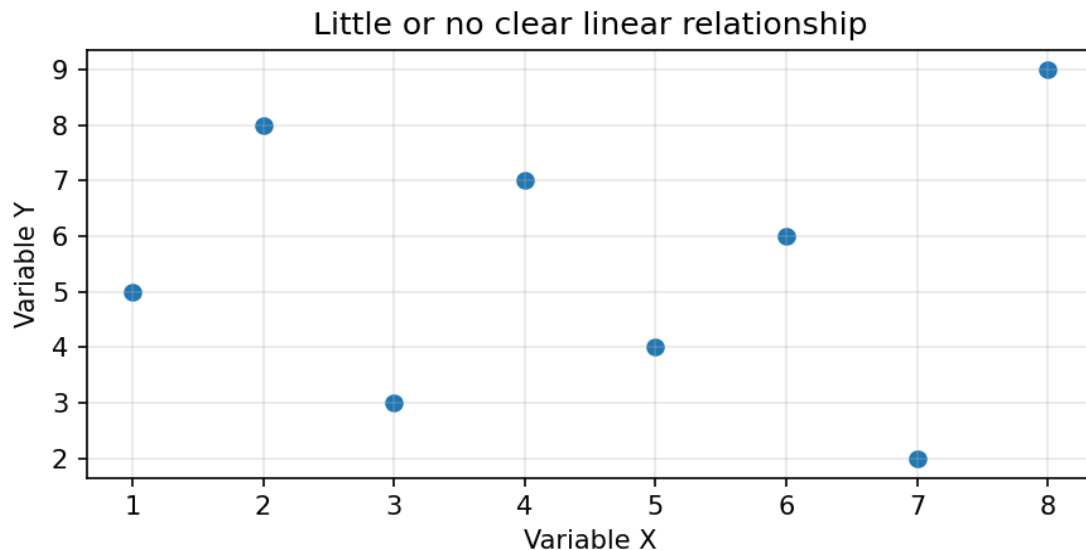
Visual guide: three common patterns



Positive: points tend to rise from left to right.



Negative: points tend to fall from left to right.



Weak/no linear relationship: no clear straight-line trend. A curved relationship can still exist even when Pearson correlation is near zero.

2. Pearson correlation coefficient

Pearson's correlation coefficient, written r , describes the strength and direction of a linear relationship. It ranges from -1 to +1.

r	General interpretation
+1.00	Perfect positive linear relationship
+0.80	Strong positive linear relationship (rough guideline)
+0.50	Moderate positive linear relationship (rough guideline)
0.00	No linear correlation
-0.50	Moderate negative linear relationship (rough guideline)
-0.80	Strong negative linear relationship (rough guideline)
-1.00	Perfect negative linear relationship

These labels are rough guidelines. Interpretation depends on context, sample size, data quality, and whether a linear model is appropriate.

Formula: $r = \text{Cov}(X, Y) / (sX \times sY)$, where Cov is sample covariance and sX and sY are sample standard deviations.

3. What is covariance?

Covariance describes whether two variables tend to vary together. Positive covariance means they tend to be above or below their own means together. Negative covariance means one tends to be above its mean when the other is below its mean. Covariance depends on measurement units and scale, while correlation is standardized and bounded between -1 and +1.

4. NumPy: calculate correlation

```
import numpy as np

study_hours = np.array([1, 2, 3, 4, 5])
marks = np.array([50, 55, 65, 70, 80])

correlation = np.corrcoef(
    study_hours,
    marks
)[0, 1]

print("Correlation:", correlation)
```

Line-by-line explanation

- 1. `import numpy as np` imports NumPy under the alias `np`.
- 2. `study_hours` stores five observations of study time.
- 3. `marks` stores the corresponding marks for the same five observations.
- 4. `np.corrcoef` calculates a 2×2 Pearson correlation matrix.

- 5. [0, 1] selects the correlation between the two variables.
- 6. print displays the coefficient.

Expected output: Correlation: 0.9934

Interpretation: this illustrative dataset has a strong positive linear association. It does not prove that study time alone causes higher marks.

5. NumPy: calculate covariance

```
covariance_matrix = np.cov(study_hours, marks)
print(covariance_matrix)
```

Sample covariance for the example is approximately 18.75. np.cov returns variances on the diagonal and covariance off the diagonal. By default it uses sample covariance (n - 1 denominator).

6. Pearson vs Spearman

Pearson uses original numerical values and measures linear association. It can be sensitive to outliers. Spearman uses ranks and measures monotonic association, so it can be useful when a relationship consistently increases or decreases but is not linear.

```
from scipy.stats import pearsonr, spearmanr

pearson_result = pearsonr(study_hours, marks)
spearman_result = spearmanr(study_hours, marks)

print("Pearson:", pearson_result.statistic)
print("Spearman:", spearman_result.statistic)
```

Both coefficients are positive for this example because values generally increase together.

7. Correlation with Pandas

```
import pandas as pd

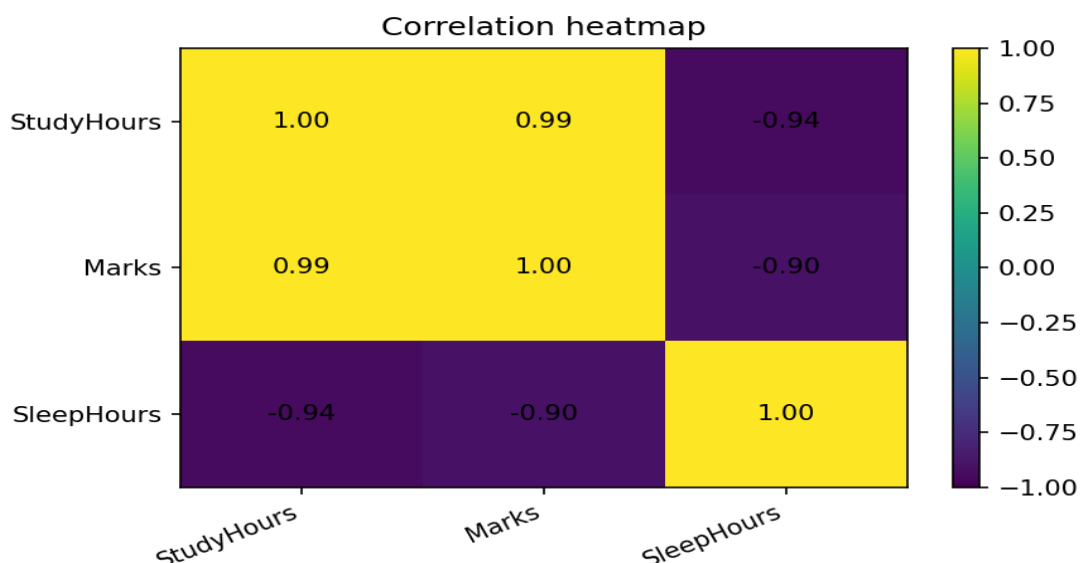
data = {
    "StudyHours": [1, 2, 3, 4, 5],
    "Marks": [50, 55, 65, 70, 80],
    "SleepHours": [8, 7, 7, 6, 6]
}
df = pd.DataFrame(data)

print(df.corr(numeric_only=True))
print(df["StudyHours"].corr(df["Marks"]))
print(df.corr(method="spearman", numeric_only=True))
```

- pd.DataFrame(data) builds a table from the dictionary.
- corr(numeric_only=True) calculates Pearson correlations between numeric columns.
- df["StudyHours"].corr(df["Marks"]) calculates one pairwise coefficient.
- method='spearman' requests rank-based correlation.

8. Scatter plots and heatmaps

Scatter plots show individual observations and can reveal trends, non-linearity, clusters, and outliers. A heatmap makes a correlation matrix easier to scan; the numbers are the coefficients, and color is a visual aid.



```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
correlation_matrix = df.corr(numeric_only=True)
sns.heatmap(correlation_matrix, annot=True,
            cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

- `annot=True` writes coefficients inside the cells.
- `cmap` selects a color palette.
- `vmin=-1` and `vmax=1` set the full correlation scale.
- `plt.tight_layout` helps prevent labels being clipped.
- `plt.show` displays the chart.

9. Correlation does not prove causation

If two variables move together, this does not prove that one causes the other. A third variable may influence both, the association may be coincidental, or the sample may be biased. For example, temperature may increase both ice-cream sales and swimming activity. Consider study design, confounders, time order, and data quality before making causal claims.

10. Practice tasks

1. Create NumPy arrays for study hours and marks, then calculate Pearson correlation.
2. Calculate sample covariance for the same arrays.
3. Create a scatter plot and describe its direction.
4. Calculate Spearman correlation using SciPy.
5. Create a DataFrame with three numeric variables and calculate its correlation matrix.
6. Generate a heatmap of the correlation matrix.
7. Explain covariance versus correlation.
8. Explain when Spearman may be more suitable than Pearson.
9. Give an example of why correlation does not prove causation.
10. Complete the Student Performance Relationship Analyzer project.

11. Complete solutions

Solution 1: Pearson correlation

```
import numpy as np
study_hours = np.array([1, 2, 3, 4, 5])
marks = np.array([50, 55, 65, 70, 80])
print(np.corrcoef(study_hours, marks)[0, 1])
```

Output: approximately 0.9934, a positive coefficient.

Solution 2: covariance

```
print(np.cov(study_hours, marks)[0, 1])
```

Output: approximately 18.75. It uses the product of the variables' units.

Solution 3: scatter plot

```
import matplotlib.pyplot as plt
plt.scatter(study_hours, marks)
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.grid(True)
plt.show()
```

Expected interpretation: points trend upward, indicating a positive association.

Solution 4: Spearman correlation

```
from scipy.stats import spearmanr
print(spearmanr(study_hours, marks).statistic)
```

Expected output: a positive value close to 1 for this example.

Solution 5: correlation matrix

```
import pandas as pd
df = pd.DataFrame({
    "StudyHours": [1, 2, 3, 4, 5],
    "Marks": [50, 55, 65, 70, 80],
    "SleepHours": [8, 7, 7, 6, 6]
})
print(df.corr(numeric_only=True))
```

The result is a 3×3 matrix. The diagonal is 1 because each variable is perfectly correlated with itself.

Solution 6: heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.heatmap(df.corr(numeric_only=True), annot=True,
            vmin=-1, vmax=1, cmap="coolwarm")
plt.show()
```

Each cell displays a pairwise Pearson coefficient.

Solution 7: covariance vs correlation

Covariance describes co-variation but depends on units and scale. Correlation is standardized, unit-free, and ranges from -1 to +1.

Solution 8: choosing Spearman

Use Spearman when the relationship is monotonic but not necessarily linear, or when ranks are more meaningful than exact distances.

Solution 9: correlation vs causation

Ice-cream sales and swimming activity may both rise in hot weather. Their association does not mean ice cream causes swimming; temperature may influence both.

Solution 10: full project

The complete project appears below with its own code explanation.

12. Full project: Student Performance Relationship Analyzer

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "Student": ["Ali", "Usman", "Hamza", "Ahmad", "Bilal",
               "Hassan", "Zain", "Omar", "Ayan", "Saad"],
    "StudyHours": [1, 2, 3, 4, 5, 2, 6, 3, 5, 4],
    "SleepHours": [8, 7, 7, 6, 7, 8, 6, 7, 6, 8],
    "Attendance": [60, 65, 70, 80, 85, 68, 95, 75, 90, 82],
    "Marks": [45, 50, 60, 70, 78, 55, 92, 65, 88, 75]
}
df = pd.DataFrame(data)
numeric_df = df.select_dtypes(include="number")
correlation_matrix = numeric_df.corr()

print("Correlation with Marks:")
print(correlation_matrix["Marks"].sort_values(ascending=False))

sns.heatmap(correlation_matrix, annot=True,
            cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Student Performance Correlation")
plt.tight_layout()
plt.show()

sns.scatterplot(data=df, x="StudyHours", y="Marks")
plt.title("Study Hours vs Marks")
plt.tight_layout()
plt.show()
```

13. Full project code explained

1. Import Pandas, Seaborn, and Matplotlib for data and plots.
2. The data dictionary stores one list per column. Lists must have equal lengths and corresponding row order.
3. `pd.DataFrame(data)` creates a table from the dictionary.
4. `select_dtypes(include='number')` excludes student names and keeps numeric columns.
5. `numeric_df.corr()` computes pairwise Pearson correlations.
6. `correlation_matrix["Marks"]` selects each numeric feature's correlation with Marks.
7. `sort_values(ascending=False)` orders those correlations from largest to smallest.
8. `sns.heatmap` draws the matrix and annotates each coefficient.
9. `plt.tight_layout` reduces label clipping.
10. `sns.scatterplot` displays individual StudyHours and Marks observations.
11. `plt.show` displays the figures in a notebook or Python plotting environment.

Expected interpretation: the printed series lists correlations with Marks. The heatmap summarizes all pairwise relationships, and the scatter plot shows study hours versus marks. Run the code to obtain exact coefficients for this ten-row illustrative dataset.

14. Revision summary

- Pearson correlation measures linear association and ranges from -1 to +1.
- Covariance describes co-variation but depends on units.
- Spearman uses ranks and measures monotonic association.
- Scatter plots help reveal trends, outliers, clusters, and non-linearity.
- Heatmaps summarize a correlation matrix.
- Correlation does not prove causation.

Day 22 completion checklist

- ■ Explain positive, negative, and near-zero linear correlation.
- ■ Calculate covariance, Pearson, and Spearman correlation.
- ■ Build a Pandas correlation matrix.
- ■ Create and interpret scatter plots and heatmaps.
- ■ Explain why correlation does not establish causation.
- ■ Run and explain the complete project.